

Stochastic programming

SAA: adaptive sampling

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Motivation

Reminder: we consider the stochastic problem

$$\min_{x \in S} g(x) = \mathbb{E}_P [G(x, \xi)],$$

where

- $x \in \mathbb{R}^m$
- S is a compact subset of $\mathbb{R}^m$
- ξ is a real random vector defined on $(\Xi, \mathcal{F}, P)$ and taking values in $(\mathbb{R}^k, \mathcal{B}^k)$ ($\mathcal{B}^k$ is the Borel measure)
- $G : \mathbb{R}^m \times \mathbb{R}^k \rightarrow \mathbb{R}$

Sample average approximation:

$$\min_{x \in S} \hat{g}_N(x) = \frac{1}{N} \sum_{i=1}^N G(x, \xi_i),$$

Convergence

In addition to our previous consistency results for $N \rightarrow \infty$, the central limit theorem tells us that, if the draws are independent and identically distributed (i.i.d.) (and finite $g(x)$),

$$\sqrt{N}[\hat{g}_N(x) - g(x)] \Rightarrow N(0, \sigma^2(x)),$$

where $\sigma^2(x) = \text{var}(G(x, \xi))$, and $\Rightarrow$ denotes convergence in distribution.

This result is only valid for a given x . It is necessary to set stronger conditions in order to have a functional convergence.

Note: under our assumptions, $\hat{g}_N(x)$ is continuous over S , and can thus be considered as a point in the Banach space $C(s)$.

Banach space $C(S)$

$C(S)$ is the space of continuous functions $\psi : S \rightarrow \mathbb{R}$, equipped with the sup-norm $\|\psi\| := \sup_{x \in S} |\psi(x)|$

$C(S)$ is a Banach space, i.e. a normed vector space, complete under the distance induced by its norm. A metric space M is said complete or complete space if every Cauchy sequence in M has a limit in M (i.e. it converges in M).

We will extend the (pointwise) central limit theorem to a functional central limit theorem, assuming as usual that the draws are i.i.d.

Assumptions

1. $\forall x \in S$, $G(x, \cdot)$ is measurable (i.e. its expectation exists).
2. $\exists \bar{x} \in S$ such that $E_P[G(\bar{x}, \xi)^2] < \infty$.
3. (Lipschitz continuity condition) $\exists K(\xi) \geq 0$ such that $E[K(\xi)]$ is finite, and $\forall x_1, x_2 \in S$ and a.e. ξ ,

$$|G(x_1, \xi) - G(x_2, \xi)| \leq K(\xi) \|x_1 - x_2\|,$$

We assume moreover that $E[K^2(\xi)] < \infty$.

Functional central limit theorem

Under these conditions,

$$N^{1/2}[\hat{g}_n - g] \Rightarrow Y \in C(S).$$

If $x_1, \dots, x_k \in S$, $(Y(x_1), \dots, Y(x_k)) \sim N(0, \Sigma)$, where Σ is the covariance matrix from $(G(x_1, \xi), \dots, G(x_k, \xi))$.

If $N^{1/2}(\hat{g}_N - g) \Rightarrow Y \in C(S)$ (with $\{\hat{g}_n\}$ and g in $C(S)$), and

$$\hat{v}_N = \min_{x \in S} \hat{g}_N(x) \text{ and } v^* = \min_{x \in S} g(x),$$

then

$$N^{1/2}(\hat{v}_N - v^*) \Rightarrow \min_{x \in S^*} Y(x).$$

Convergence of global solutions

If S^* is a singleton, under the previous assumptions, in the i.i.d. case,

$$N^{1/2}(\hat{v}_n - v^*) \Rightarrow N(0, \sigma^2(x^*)).$$

Under some additional conditions, we also have the convergence of $E[\hat{v}_N]$ to v^* .

But all these results become difficult to extend in the case of local optimization.

However, we see that the results should be better with larger N , but at a higher computational cost, as

$$\hat{g}_N(x) = \frac{1}{N} \sum_{i=1}^N G(x, \xi_i).$$

External adaptive method

What interests us?

$$\min_{x \in S} \hat{g}_N(x) = \frac{1}{N} \sum_{i=1}^N G(x, \xi_i).$$

We can start with a small sample and extend it over the iterations: the adaptive sampling procedure can be **external** to the algorithm, or **internal**.

An external approach is known as **retrospective approximation** consists in repeatedly applying the optimization algorithm with samples of increasing sizes.

Retrospective approximation (RA)

Introduced by Healy and Schruben in 1991.

Let k be the iteration index. Components:

1. A procedure for solving a generated sample-path problem to specified tolerance vector ϵ_k , delivering a solution x_k .
2. A sequence $\{N_k\}$ of sample sizes tending to infinity.
3. A sequence $\{\epsilon_k\}$ of error-tolerances tending to zero.
4. A sequence of weights $\{w_{kj}, j = 1, 2, \dots, k\}$ for each iteration.

We define

$$\bar{X}_k := \sum_{j=1}^k w_{kj} X_j.$$

Retrospective approximation: principle

- Step 0. Set $k = 1$ and choose $\bar{x}_0 = x_0 \in S$.
- Step 1. Generate a sample-path problem with sample size N_k , with a “warm start”, i.e. starting from $\bar{x}_{k-1}$ to solve the generated problem with the error-tolerance ϵ_k . Denote the obtained retrospective solution by x_k .
- Step 2. Compute the solution $\bar{x}_k$.
- Step 3. Set $k \leftarrow k + 1$ and go to Step 1.

Retrospective approximation: assumptions

- The true function g has a unique minimizer $x^* \in S$.
- $G(\cdot, \xi)$ is Lipschitz with Lipschitz constant $L(\xi)$ on S a.s., and $\mathbb{E}[L(\xi)] < \infty$.
- $G(\cdot, \xi)$ is continuously differentiable at any $x \in \mathcal{B}(x^*, \epsilon)$, $\epsilon > 0$, a.s.
- $\exists x \in S$ such that $\mathbb{E}[\|\nabla_x G(x, \xi)\|^2] < \infty$.
- The sample function $\hat{g}_N(x)$ has a unique minimum x_N^* a.s.
- When $\hat{g}_N(x)$ attains a unique minimum x_N^* , $\hat{g}_N(x)$ is twice differentiable at x_N^* . Furthermore, the $\nabla^2 \hat{g}_N(x_N^*)$ is positive definite with smallest eigenvalue uniformly bounded away from 0 a.s.
- The solution x_k obtained from the k^{th} iteration of RA satisfies $\|\nabla \hat{g}_{N_k}(x_k)\| \leq \epsilon_k$.

Retrospective approximation: assumptions

- The numerical procedure used to solve the sample-path problems in RA exhibits p^{th} -order sublinear convergence or p^{th} -order linear convergence with respect to the observed derivatives.
- The sample sizes are increased linearly, i.e., $N_k = cN_{k-1} > 1$ for all k .
- The error-tolerances are chosen so that $\epsilon_k = O(1/\sqrt{N_k})$.

Retrospective approximation: convergence rate

Under the previous assumptions,

$$C_k \|x_k - x^*\|^2 = O_p(1),$$

as $k \rightarrow \infty$, where C_k is the total amount of computational work done until the k^{th} iteration and is given by $C_k = \sum_{i=1}^k Q_i N_i$. Here Q_i is the number of points visited by the numerical procedure during the i th iteration.

We recover the convergence rate of stochastic approximation method.

Internal-external adaptive method

- The major issue with this procedure is how to quantify the word “approximative” in Step 1. If no care is taken, the resulting algorithm can in fact be more time-consuming than the direct minimization of $\hat{g}_{N_{\max}}$.
- We can also replace the stopping test on $N_{\max}$ by a test of the criticality conditions of optimality.
- The **internal approach** is a non-monotone strategy that depends on the underlying optimization methods. Here, we consider the unconstrained case.
- More precisely, we generate a sample before the optimization process, with $N_{\max}$ i.i.d. random draws. At iteration k , we will use a subset of this initial sample, using N_k of the $N_{\max}$ random draws, typically the first ones.

Accuracy estimation

- This implies that $\hat{g}_N$ is a smooth function, well defined for each choice of N .
- Let α_δ be the quantile of a $\mathcal{N}(0, 1)$ associated to some significance level δ , i.e.

$$P_\xi[-\alpha_\delta \leq Y \leq \alpha_\delta] = \delta,$$

where $Y \sim \mathcal{N}(0, 1)$.

- Central limit theorem (background notes, where the quantile is written $\Phi^{-1}(1 - \delta/2)$):

$$g(x) - \hat{g}_N(x) \Rightarrow \mathcal{N}\left(0, \frac{\sigma^2(x)}{N}\right),$$

- Confidence interval for $g(x)$ around $\hat{g}_N(x)$:

$$[\hat{g}_N(x) - \epsilon_N^\delta(x), \hat{g}_N(x) + \epsilon_N^\delta(x)],$$

Accuracy estimation (cont'd)

$\epsilon_N^\delta(x)$ is given by

$$\epsilon_N^\delta(x) = \alpha_\delta \frac{\sigma(x)}{\sqrt{N}}.$$

Typically, we will choose $\alpha_{0.9} \approx 1.64$ or $\alpha_{0.95} \approx 1.96$.

In practice, we do not know $\sigma^2(x)$, but we can use its estimator

$$\hat{\sigma}_N^2(x) = \frac{1}{N-1} \sum_{i=1}^N (G(x, \xi_i) - \hat{g}_N(x))^2.$$

We will exploit this error estimation in the context of trust-region methods.

Basic principles

- Basic idea: if the model approximates the objective function well enough w.r.t. the accuracy of the objective function (which depends on the sample size), we presume that we could work with a less accurate approximation, and reduce the sample size.
- If the adequacy of the model w.r.t. the accuracy of the objective function is poor, we can increase the sample size in an attempt to correct this deficiency.
- We assume the assumptions developed for the consistency analysis hold.

Algorithm BTRDA (Bastin, Cirillo, Toint, 2006)

(Basic) trust-region algorithm with dynamic accuracy.

Step 0. Initialization. initial point: x_0 , initial trust-region radius: Δ_0 . Set η_1 and η_2 such that $0 < \eta_1 \leq \eta_2 < 1$ (for instance, $\eta_1 = 0.01$ and $\eta_2 = 0.75$), $N_{\min} = N_{\min}^0$ and N_0 satisfying $\|\nabla \hat{g}_{N_0}(x_0)\| \neq 0$ if $\epsilon_{\delta}^{N_0}(x_{k+1}) \neq 0$, except if $N_0 = N_{\max}$. Compute $\hat{g}_{N_0}(x_0)$ and set $k = 0$, $t = 0$.

Step 1. Stopping test. Stop if $\|\nabla \hat{g}_{N_k}(x_k)\| = 0$ and either $N_k = N_{\max}$, either $\epsilon_{\delta}^{N_k}(x_k) = 0$. Otherwise, go to Step 2.

Step 2. Model definition Define a model $m_k^{N_k}$ of $\hat{g}_{N_k}(x)$ in $\mathcal{B}_k$. Compute a new adequate sample size N^+ , and set $N^- = N_k$.

Algorithm BTRDA (cont'd)

Step 3. Step computation Compute a step s_k that sufficiently reduces $m_k^{N_k}$ and s.t. $x_k + s_k \in \mathcal{B}_k$. Set

$$\Delta m_k^{N_k} = m_k^{N_k}(x_k) - m_k^{N_k}(x_k + s_k).$$

Step 4. Comparison of decreases Compute $\hat{g}_{N^+}(x_k + s_k)$ and

$$\rho_k = \frac{\hat{g}_{N_k}(x_k) - \hat{g}_{N^+}(x_k + s_k)}{\Delta m_k^{N_k}}.$$

Step 5. Sample size update If $\rho_k < \eta_1$ and $N_k \neq N^+$, modify N^- or the candidate sample size N^+ in order to take account of variance differences. Update ρ_k .

Algorithm BTRDA (cont'd)

Step 6. Candidate iterate acceptance If $\rho_k < \eta_1$, define $x_{k+1} = x_k$, $N_{k+1} = N^-$. Otherwise, define $x_{k+1} = x_k + s_k$ and set $N_{k+1} = N^+$; increment t .

If $N_{k+1} \neq N^{\max}$, $\|\nabla \hat{g}_{N_{k+1}}(x_{k+1})\| = 0$, and $\epsilon_{\delta}^{N_{k+1}}(x_{k+1}) \neq 0$, increase N_{k+1} to some size less or equal to $N_{\max}$ such that $\|\nabla \hat{g}_{N_{k+1}}(x_{k+1})\| \neq 0$ if $N_{k+1} \neq N_{\max}$, and compute $\hat{g}_{N_{k+1}}(x_{k+1})$.

If $N_k = N_{k+1}$ or if a sufficient decrease has been observed since the last evaluation of $\hat{g}_{N_{k+1}}$, set $N_{\min}^{k+1} = N_{\min}^k$. Otherwise, set $N_{\min}^{k+1} > N_{\min}^k$.

Algorithm: BTRDA (cont'd)

Step 7. Trust region radius update

$$\Delta_{k+1} \in \begin{cases} [\Delta_k, \infty) & \text{if } \rho_k \geq \eta_2, \\ [\gamma_2 \Delta_k, \Delta_k] & \text{if } \rho_k \in [\eta_1, \eta_2), \\ [\gamma_1 \Delta_k, \gamma_2 \Delta_k] & \text{if } \rho_k < \eta_1, \end{cases}$$

In this algorithm the variable t is used to count the number of successful iterations.

Note: the algorithms BTR and BTRDA coincide if we fix N_k to $N_{\max}$ for all $k \geq 0$.

Variable sample size strategy

- Before the optimization, the user chooses a maximal sample size $N_{\max}$ (for instance the number of observations).
- A minimum sample size $N_{\min}^0$.
- Define $N_0 = \max\{N_{\min}^0, 0.1 N_{\max}\}$ if $\|\nabla \hat{g}_{N_0}(x_0)\| \neq 0$ and $\epsilon_{\delta}^{N_0}(x_0) \neq 0$, $N_0 = N_{\max}$ otherwise.

Choice of N^+ in (Step 3)

Sample size required to obtain an accuracy equal to Δm_k :

$$N^s = \max \left\{ N_{\min}^k, \left\lceil \frac{\alpha_\delta^2 \hat{\sigma}_N^2(x)}{(\Delta m_k^{N_k})^2} \right\rceil \right\}.$$

Set

$$\tau_1^k = \frac{\Delta m_k^{N_k}}{\epsilon_\delta^{N_k}(x_k)}, \quad \tau_2^k = \frac{N_k}{\min\{N_{\max}, N^s\}}.$$

Set $N^+ = \max\{N', N_{\min}^k\}$, where

$$N' = \begin{cases} \min\{\lceil \chi_1 N_{\max} \rceil, \lceil N^s \rceil\} & \text{if } \tau_1^k \geq 1, \\ \min\{\lceil \chi_1 N_{\max} \rceil, \lceil \tau_1^k N^s \rceil\} & \text{if } \tau_1^k < 1 \text{ and } \tau_1^k \geq \tau_2^k, \\ \lceil \chi_1 N_{\max} \rceil & \text{if } \nu_1 \leq \tau_1^k < 1 \text{ and } \tau_1^k < \tau_2^k, \\ N_{\max} & \text{if } \tau_1^k < \nu_1 \text{ and } \tau_1^k < \tau_2^k. \end{cases}$$

Variable sample size strategy (cont'd)

A possible value for χ_1 is 0.5.

- $\tau_1^k \geq 1$: $\Delta m_k \geq$ estimated accuracy.
- $\tau_1^k < 1$: $\Delta m_k < \text{accuracy}$. However, a sufficient improvement during several consecutive iterations can lead to a significant decrease.
 - If $\tau_1^k \geq \tau_2^k$, compute a sample size smaller than N^s , such that an improvement of the order $\epsilon_\delta^{N_k}(z_k)$ would be reached in approximately $\lceil \tau_1^k \rceil$ iterations if τ_1^j is similar to τ_1^k for j close to k .
 - If $\tau_1^k < \tau_2^k$, it can nevertheless be cheaper to continue to work with a smaller sample size, defined again as $\lceil \chi_1 N_{\max} \rceil$, as long as τ_1^k is greater than some threshold ν_1 (for instance 0.2).

Accuracy differences

If N^+ is not equal to N_k , the computation of

$$\hat{g}_{N_k}(x_k) - \hat{g}_{N^+}(x_k + s_k)$$

is affected by the change in approximation variance. This can lead to $\rho_k < \eta_1$, even if the model $m_k^{N_k}$ gives a good prediction for the sample size N^k .

If $N^+ > N_k$, compute $\hat{g}_{N^+}(x_k)$, $\Delta m_k^{N^+}$ and $\epsilon_\delta^{N^+}(x_k)$, otherwise if $N^+ < N_k$ compute $\hat{g}_{N_k}(x_k + s_k)$. Set N^- to $\max\{N_k, N^+\}$, and

$$\rho_k = \frac{\hat{g}_{N^-}(x_k) - \hat{g}_{N^-}(x_k + s_k)}{\Delta m_k^{N^-}}.$$

Convergence

Additional computational safeguards allow to establish the following result.

Theorem

Under some regularity assumptions, if

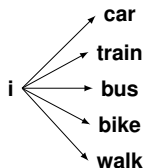
$$\exists \kappa > 0 \text{ such that } \epsilon_{\delta}^{N_k}(x_k) \geq \kappa,$$

for all k large enough, then, a.s., the algorithm converges with $N_i = N_{\max}$ for all i large enough.

Application to discrete choice theory

Discrete **set of alternatives** available for individual i : $\mathcal{A}(i)$.

Example: Which mode of transport have you chosen to come?



Each alternative has some utility.

Utility U_{ij} of $A_j \in \mathcal{A}(i)$: $U_{ij} = V(\beta) + \epsilon_{ij}$.

β : parameters to be estimated.

rational individuals: a person chooses the alternative of **maximum utility**

→ choice of A_j if $U_{ij} \geq U_{in}, \forall A_n \in \mathcal{A}(i)$.

Important case: Gumbel distributed residuals ϵ_{ij} (mean 0, scale factor μ): → **multinomial logit** (MNL).

Probability that individual i choose A_j :

$$P_{ij} = \frac{e^{\mu V_{ij}(\beta)}}{\sum_{n=1}^N e^{\mu V_{in}(\beta)}}$$

Mixed Logit models

Allow **heterogeneity** in parameters inside the population.

$$\beta = \beta(\gamma, \theta),$$

γ : **random vector**, e.g. vector of independent $N(0, 1)$;

θ : **vector of parameters**, e.g. vector of means and std dev.

Probability choice of A_j by individual i :

$$P_{ij}(\theta) = E_{\xi} [L_{ij}(\gamma, \theta)] = \int L_{ij}(\gamma, \theta) f(\gamma) d\gamma$$

This integral is approximated by

$$SP_{ij}^R = \frac{1}{R} \sum_{r=1}^R L_{ij}(\gamma_r, \theta)$$

SAA problem:

$$\max_{\theta} \hat{g}_R(\theta) = \max_{\theta} SLL(\theta) = \max_{\theta} \frac{1}{N} \sum_{n=1}^N \ln SP_{ij}^R$$

→ suggests to consider **stochastic programming** techniques.

Properties of Mixed Logit

Stochastic programming:

$$\min_z g(z) = \min_z E_{\xi} [G(z, \xi)]$$

Mixed logit:

$$\max_{\theta} g(\theta) = \max_{\theta} LL(\theta) = \frac{1}{n} \sum_n \ln E_{\xi} [L_{in}(\gamma, \theta)]$$

Consistency properties can be adapted. Assume I fixed and R grows toward ∞ .

If θ_R^* , $R = 1, \dots$, are first (second)-order critical for the corresponding SAA problem, under some regularity conditions, for almost every sequence of random draws, there exists some limit point θ^* of $(\theta_R^*)_{R=1}^{\infty}$ that is first (second)-order critical.

Error estimation

With an i.i.d. sample for each individual, we have, from the delta method (see for instance Shapiro and Rubinstein):

$$LL(\theta) - SLL^R(\theta) \Rightarrow N\left(0, \frac{1}{I} \sqrt{\sum_{i=1}^I \frac{\sigma_{ij_i}^2(\theta)}{R(P_{ij_i}(\theta))^2}}\right)$$

Asymptotic value of the confidence interval radius:

$$\epsilon_\delta = \alpha_\delta \frac{1}{I} \sqrt{\sum_{i=1}^I \frac{\sigma_{ij_i}^2(\theta)}{R(P_{ij_i}(\theta))^2}}$$

δ : significance level; $\alpha_{0.9} \approx 1.65$.

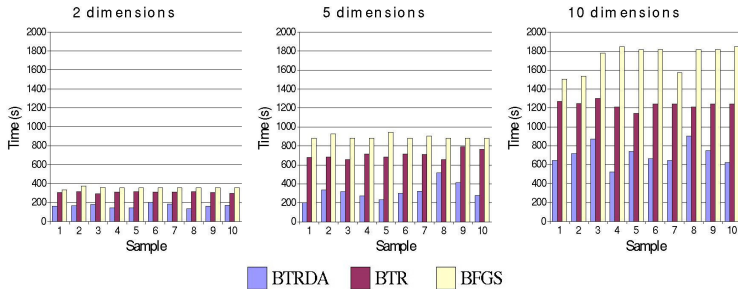
Bias of simulation (Taylor expansion):

$$B := E[SLL^R(\theta)] - LL(\theta) = -\frac{I\epsilon_\delta^2}{2\alpha_\delta^2}$$

In practice, use of SAA estimators $\sigma_{ij_i}^R(\theta)$ and $P_{ij_i}^R(\theta)$.

Algorithms comparison

5000 individuals, 5 alternatives, coefficients $\sim N(0.5, 1)$



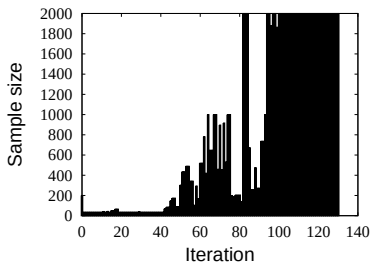
	2 dimensions	5 dimensions	10 dimensions
BTRDA	167s	319s	709s
BTR	310s	706s	1235s
BFGS	357s	894s	1736s

Results obtained with a Pentium IV, 2 Ghz. BFGS method has been implemented using the More-Thuente linesearch.

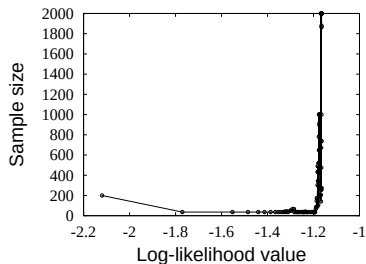
Example

Mode choice model: **Mobidrive** data (Axhausen and al.)

- 5799 observations;
- 5 alternatives;
- 14 parameters (3 random, normally distributed).



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Starting value



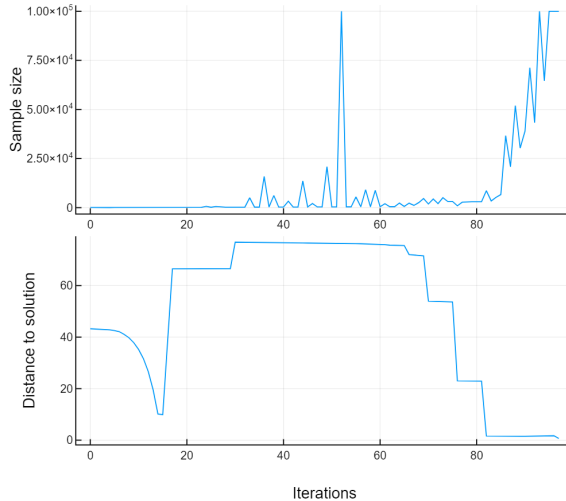
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Maximum value

Possible improvement: how to draw samples?

Sampling methods are notorious for generalization issues. Common strategies:

- *Independent Random Numbers* (IRN) (e.g. stochastic gradient descent)
- Incremental *Common Random Numbers* (CRN): take the first N_K numbers from a predefined sequence of random draws: $\mathcal{N}_K = \{\xi_1, \dots, \xi_{N_K}\}$

Incremental Common Random Numbers



IRN/CRN combination (ICRN)

1. $N_{k+1} > N_k$

Draw new samples $\xi_{N_k+1, \dots, N_{k+1}}$ and set

$$\mathcal{N}_{k+1} = \mathcal{N}_k \cup \left\{ \xi_{N_k+1, \dots, N_{k+1}} \right\}.$$

2. $N_{k+1} < N_k$

$$\mathcal{N}_{k+1} = \mathcal{N}_k \setminus \{\text{uniformly randomly selected elements in } \mathcal{N}_k\}$$

This heuristic reduces the risk to be trapped in a local minimizer or a saddle point.